

Gate 7A - Final Job-Ready Practical

Protected first attempt | Cold-chain distribution | Conditional RC1

Scenario

A cold-chain distributor ships temperature-sensitive medical products through regional depots. You receive shipment files, paginated temperature-alert API events and an operational depot database. Build a trustworthy production-style data platform that identifies delivery state and temperature excursions without losing auditability.

Required work

- Read requirements and source contracts; declare grain, business keys and freshness expectations before coding.
- Ingest the supplied CSV + paginated API + operational database source and preserve raw evidence.
- Implement critical validation, quarantine and reconciliation. Bad data must not silently reach trusted output.
- Build a relational/analytics model with clear facts/dimensions or equivalent trusted grain.
- Use dbt-style dependency/tests or equivalent tested transformation evidence.
- Orchestrate the repeatable path with failure-safe dependencies, retries and bounded backfill/reprocess logic.
- Use PySpark/Delta only where the scale requirement justifies it; document partition/performance reasoning.
- Design the Microsoft Fabric/OneLake/Lakehouse deployment, access model, CI/CD and monitoring.
- Prove safe rerun/idempotency and explain one deliberate failure/recovery path.
- Commit/document the work as a reproducible engineering repository and complete a technical defense.
- Run the offline Gate checker and save the output. Live runtime evidence remains separate before FINAL.

Critical incidents

- Duplicate shipment file must not double count.
- One shipment has a missing business key.
- One temperature event is malformed.
- A late temperature alert arrives after the normal watermark.
- A transformation quality test must be able to block publish.
- One simulated deployment/pipeline failure must have a safe recovery path.

Protected rules

- Documentation is allowed.
- No direct AI solution until first attempt and checker output are saved.
- Do not open Gate 7A Review before saving the first attempt.
- Do not use real credentials or real customer/company data.